THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 6: **Triangles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. All squares are: 1

(a) congruent **(b)** similar **(c)** neither **(d)** equal in area

2. In $\triangle ABC$, $DE \parallel BC$ with $AD = 4$ cm, $DB = 6$ cm and $AE = 5$ cm. Then EC is: 1

(a) 7.5 cm **(b)** 6 cm **(c)** 8 cm **(d)** 5 cm

3. $\triangle ABC \sim \triangle DEF$ with $\frac{AB}{DE} = \frac{2}{3}$. If the perimeter of $\triangle ABC$ is 24 cm, the perimeter of $\triangle DEF$ is: 1

(a) 16 cm **(b)** 36 cm **(c)** 48 cm **(d)** 54 cm

4. A student states: "Two triangles are similar if two sides of one are proportional to two sides of the other." The correct response is: 1

(a) Correct
(b) Incorrect — the included angles must also be equal (SAS)
(c) Correct only for right triangles
(d) Cannot be determined

5. If $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$, then $\triangle ABC \sim \triangle DEF$ by: 1

(a) AA **(b)** SAS **(c)** SSS **(d)** ASA

6. If $\triangle PQR \sim \triangle XYZ$ with $\angle Q = 50^\circ$ and $\angle R = 70^\circ$, then $\angle X$ is: 1

(a) 50° **(b)** 60° **(c)** 70° **(d)** 120°

7. In $\triangle ABC$, $AD = 3$, $DB = 4$, $AE = 4.5$ and $EC = 6$ (D on AB, E on AC). Student A says $DE \parallel BC$; Student B says it is not. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Cannot be decided

8. A 6 m pole casts a 4 m shadow. At the same time a building casts an 18 m shadow. The height of the building is: 1

(a) 27 m **(b)** 24 m **(c)** 12 m **(d)** 30 m

9. In $\triangle ABC$, $DE \parallel BC$, $\frac{AD}{AB} = \frac{2}{5}$ and $DE = 4$ cm. Then BC is: 1

(a) 8 cm **(b)** 10 cm **(c)** 6 cm **(d)** 12 cm

10. Similar figures always have: 1

(a) the same size
(b) the same shape, not necessarily the same size
(c) equal areas
(d) equal perimeters

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, $DE \parallel BC$ where D is on AB and E is on AC. If $AD = 6$ cm, $AE = 8$ cm and $EC = 4$ cm, find DB. **2**

12. State the AA similarity criterion. Are two right triangles, each having an acute angle of 35°, similar? Why? **2**

13. In $\triangle ABC$, $DE \parallel BC$ with $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15$ cm. Find AE. **2**

14. A student claims that $\triangle ABC$ with $\angle A = 50^\circ, \angle B = 60^\circ$ and $\triangle PQR$ with $\angle P = 50^\circ, \angle R = 70^\circ$ are not similar. Justify or refute. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. E and F are points on sides PQ and PR of $\triangle PQR$ with $PQ = 1.28$ cm, $PR = 2.56$ cm, $PE = 0.18$ cm and $PF = 0.36$ cm. Is $EF \parallel QR$? Justify. **3**

16. ABCD is a trapezium with $AB \parallel DC$. E and F are points on AD and BC such that $EF \parallel AB$. Show that $\frac{AE}{ED} = \frac{BF}{FC}$. **3**

17. D is a point on side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Show that $CA^2 = CB \cdot CD$. **3**

OR

In $\triangle ABC$, altitudes AD and CE intersect each other at P. Show that $\triangle ABD \sim \triangle CBE$. **3**

18. A 1.8 m tall man casts a shadow 1.2 m long, and at the same time a tower casts a shadow 24 m long. Find the height of the tower. What will be the length of the shadow of a 9 m pole at that time? **3**

OR

In $\triangle ABC$, $DE \parallel BC$ with $AD = x + 1$, $DB = 3x - 1$, $AE = x + 3$ and $EC = 3x + 4$. Find x. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that if a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. **5**

OR

E is a point on side AD produced of a parallelogram ABCD and BE intersects CD at F. **5**
(i) Show that $\triangle ABE \sim \triangle CFB$.
(ii) If $AB = 8$ cm, $BC = 9$ cm and $AE = 12$ cm, find CF.

20. **Case study:** A lamp is 4.8 m above the ground. A boy 1.6 m tall walks away from the base of the lamp-post at a speed of 1.5 m/s. **5**
(i) Which similarity criterion applies?
(ii) Find the length of his shadow after 4 seconds.
(iii) After how many seconds will his shadow be 4.5 m long?

OR

If AD and PM are medians of triangles ABC and PQR respectively, where $\triangle ABC \sim \triangle PQR$, prove that $\frac{AB}{PQ} = \frac{AD}{PM}$. **5**

END OF QUESTION PAPER — **Total: 40 Marks**